离散 Lec 10. 交换环中

associate 相伴 $\exists$ unit c $ca=b$

unit a is unit $\exists a^{-1} aa^{-1}=1$

$a|b \quad \exists c \quad ac=b$

$a\sim b \to a|b \wedge b|a$

Communicative Ring → Integral Domain → Unique Factorization Domain → Principal Ided Domain → Euclideon Domain → Field

$a|b \wedge b|a \to a\sim b$ in Integral Domain : every $r\neq 0$ has unique factorization

$ca=b \quad cd\cancel{b}=\cancel{a}b \quad a=bd$

→ { 因子链 $\neg\exists a_1,a_2\dots$ s.t. $a_{t+1}|a_t \quad a_{t+1}\not\sim a_t$
素性 p is irreducible → p is prime

Unique Factorization in $\mathbb{N}$

any $n\neq 0$ $\exists$ unique $p_1\cdots p_t$ s.t. $n=p_1\cdots p_t$

any $r\in R\setminus\{0\}$

$\exists$ unique $\underbrace{a_1\dots a_t}_{\text{irreducible}}$

s.t. $r=\underbrace{u}_{\text{unit}}a_1\cdots a_t$

a is irreducible $(p|a \to p\sim a$ or $p\sim 1) \wedge a\sim 1$

$\Uparrow$

p is prime if $p|ab \to p|a$ or $p|b$

不是唯一分解整环

都是不可约. 但不对应

$6=2\cdot 3=(1-\sqrt{-5})(1+\sqrt{-5})$

$\mathbb{Z}[\sqrt{10}]=\{a+\sqrt{10}b, a,b\in\mathbb{Z}\}$ $\mathbb{Z}[\sqrt{5}]=\{a+\sqrt{-5}b \mid a,b\in\mathbb{Z}\}$

$R'\supseteq R \quad u\in R'$ $R[u]$：包含 R 和 u 的 R' 的最小的子环

$=\{r_0+r_1u+r_2u^2+\dots$

找模更小的因子.

$||a+b\sqrt{10}|| = ||a^2-10b^2||$

$||a+b\sqrt{10}|| \cdot ||c+d\sqrt{10}||$ $||(a+b\sqrt{10})(c+d\sqrt{10})||$

in UFD irre → prime

Pf. $p|ab$

$decomp(a) = p_1p_2\cdots p_t$ $decomp(b) = q_1\cdots q_s$ $= p\cdot decomp(\frac{ab}{p})$

Pf. 假设元素有两不同分解

$p_1\cdots p_t = q_1\cdots q_s$

$p_1|q_1\cdots q_s$

w.l.o.g. $p_1\sim q_1$

$p_2\cdots p_t \sim q_2\cdots q_s$

Princepal Ideal Domain (every Ideal is a princepal ideal)

满足因子链 + 素

$a_1, a_2\cdots a_3\cdots$

$a_{t+1}|a_t$

$(a_{t+1})\supseteq(a_t)$ 考虑 $I=\bigcup_i(a_i)=(b)\ni b$

$\exists i,\ b\in(a_i)$ $(a_i)=(b)=(a_{i+1})$ 相等. 不满足(真)因子链

p is irreducible

plab

~~(ab)~~ = (s)

cap)

s|p s∼p or s∼1

↓ ↓

p|a ~~s~~ 1 = αa + βp

(bp) = (t) p|t ∎ 1 = α'b + β'p

(1 - βp)(1 - β'p) = αα'ab

1 + (· · ·)p = 得 p|1

Euclideon Domain ∃ ‖ ‖ norm $R \to \mathbb{N}$ s.t.

‖a‖ = 0 ↔ a = 0 ∀a ≠ 0, b ∃ q, r, b = aq + r s.t. ‖r‖ < ‖a‖

F[x] ‖f(x)‖ = degree + 1

f(x) ≠ 0 g(x) = q(x)f(x) + r(x)

ED → PID

Pf. Ideal I ≠ {0}

Pick a ∈ I of \{0} with smallest norm

$a = \underset{a \in I \setminus \{0\}}{\operatorname{argmin}} \|a\|$

∀ b ∈ I → r ∈ I

b = qa + r ‖r‖ < ‖a‖ → b ∈ (a)

I maximum ideal → R/I Field I prime ideal ↔ integral Domain

In PID a irreducible ↔ (a) maximal

(a) ⊂ I ⊂ R

(1) (b) b|a ∧ b ∤ a

if p(x) ∈ F[x] is irreducible

F(x) / (p(x)) is a field

$\mathbb{Z}[\sqrt{10}] \cong \mathbb{Z}[x]/x^2 - 10$

$\mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}[x]/x^2 + 5$

Integral Domain R

Frac(R) = (R × R\{0}) / ∼

分式域

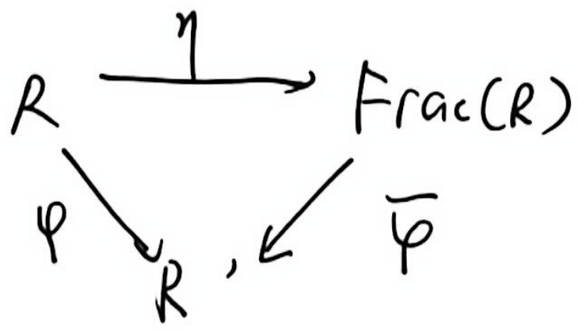
fraction field

any injective homomorphism φ: R → R'

s.t. ∀ x ∈ R\{0} φ(x) is an unit

$\overline{\varphi}(a, b) = \frac{\varphi(a)}{\varphi(b)}$

..a/b..



下CE　　F、E are fields

① $\mathbb{F}[x]/P(x) = E$　　② $E = Frac(\mathbb{F}[x])$

E is a F-linear space

characteristic 特征 $\underbrace{1+1+\cdots+1}_{char(R)} = 0$

if $p = char(R) \neq 0$，所有1相加得0元素得到子环 $\{1+1+\cdots+1\} \subseteq R$
$\cong \mathbb{Z}/p\mathbb{Z}$

否 $char(R) = 0$　$\mathbb{Z} \cong \{\frac{1+1+\cdots+1}{k} \mid k \in \mathbb{Z}\} \subseteq R$

$p = char(\mathbb{F})$

$GF(p) = \mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z} \cong K \subseteq F$
$char(F)=0$

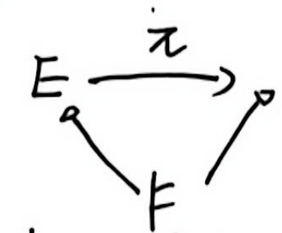
$Q = Frac(\mathbb{Z}) \cong K \subseteq F$

Field of finite size
→ prime power size
$\mathbb{F}_p[x]/(f(x))$ deg-k irreducible

$[E:F]$ = dimension of F-linear space

f is sdvable ↔ reducible in E $f(x)$

$y-x \mid f(y)$
$x \in E$



assume $f(u) = 0$ in E'

$\pi : E \to E'$　$f \mapsto f(u)$　$\alpha \mapsto \alpha(u)$

$\alpha' \in \alpha + (f(x))$
$\alpha + qt$
$\pi(\alpha') = \alpha(u) + q(u)f(u) = \pi(\alpha)$

~~eye~~ every f in $\mathbb{F}[x]$ splits in an field extension $E \supseteq F$
splits 能够完全分解
↓
spliting field
可split的最小的域